

A question about two rows becomes a join

A semantic join tests a natural-language condition with a model.
It keeps the pairs for which the model returns True.
Here the condition is: “Do these records describe the same person?”

Record a from table A

Name	Amina Patel
Date of birth	1988-04-12
Postcode	3000

Record b from table B

Name	Amina Patel
Date of birth	1988-04-12
Postcode	3000

A teaching example with constructed records and stipulated probabilities.

Fellegi–Sunter / Splink

Compare fields, then combine their evidence.

What do we observe?

Name	agrees
Date of birth	agrees
Postcode	agrees

The comparison vector is
 $\gamma = (\text{agree}, \text{agree}, \text{agree})$.

What does agreement tell us?

Suppose the name agrees in
85 of 100 same-person pairs ($m = 0.85$),
1 of 100 different-person pairs ($u = 0.01$).
Name agreement is $m/u = 85$ times as likely
under the same-person model.
It multiplies the prior odds by **85**.

Use all three comparisons and the prior

Evidence factors: name 85, birth date 450,
postcode 20.
Before comparing fields, suppose 1 in 10,000
pairs is a match: $\pi = 1/10,000$.

$$\underbrace{\frac{1}{9999}}_{\text{prior odds}} \times \underbrace{85 \times 450 \times 20}_{\text{comparison evidence}}$$

$\Rightarrow$ **98.7% posterior match probability**

This calculation assumes comparisons are
conditionally independent within same-person
pairs and within different-person pairs.

A chosen threshold/review policy determines
whether the scored pair is accepted.

Semantic join

Give both rows and the condition to an LLM.

What do we ask?

“Do these records describe
the same person?”

Supply the two Amina Patel rows above.

Fix the model, prompt and parsing rules:
together they define the
reference procedure.

What comes back?

A yes/no answer (True or False) for this pair.

If True keep the pair (a, b)

If False omit the pair (a, b)

This is a conditional illustration,
not a recorded model decision.

Apply the question across the two tables

The direct reference evaluates
every pair in $A \times B$.

The join returns the pairs
whose answer is True:

Left record	Right record
a	b (only if True)

With this identity question, retained pairs are
proposed identity links.

A Boolean answer does
not supply a calibrated
match probability.

What LOTUS and Featurized-Decomposition Join add

LOTUS: make the question a table operation

Write a condition, compose table operators
and optimize their execution.

The reference defines the intended result.
Optional optimizations reduce execution cost
under statistical accuracy targets relative to it.

FDJ: reuse work across many pairs

Derive reusable features
for individual records



cheap comparisons screen candidate pairs



reference-LLM refinement
produces the result.

Check the identity links against independent identity evidence.

Agreement with a reference LLM measures execution
fidelity; it does not establish who is the same person.

Another condition, such as “Does this report mention this medicine?”, defines a different relation
and needs its own evaluation.